

USABILITY EVALUATION

Usability Evaluation of AMD Software: Adrenalin Edition

Chinmoy Guha

ID 22-48056-2 · Section A

American International University-Bangladesh
Department of Computer Science

AMD

UX

HUMAN-COMPUTER
INTERACTION

Five Norman concepts frame the evaluation

01 Affordance

02 Visibility

03 Mapping

04 System Image

05 Gulfs

EVALUATION SCOPE

AMD Software 24.9.1



01

DISCOVERING ACTION

Affordance

Does the interface suggest how to act?

What affordance means

DEFINITION

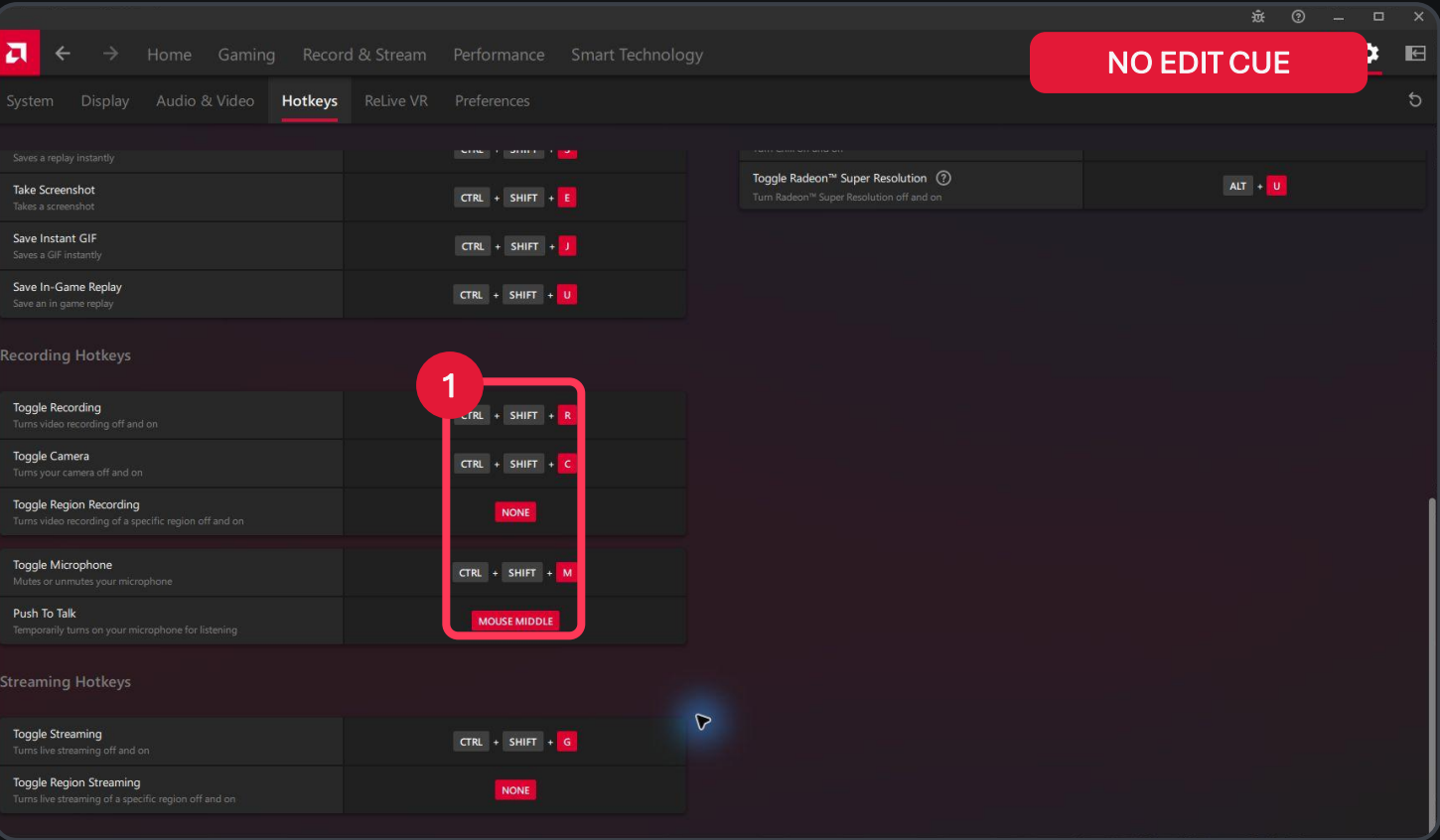
Affordance is the action possibility a user perceives from an interface object. Signifiers such as labels, icons, shape and feedback communicate how to perform that action.

In interfaces, good signifiers make the intended affordance visible.

EVALUATION QUESTIONS

- 1 What action does the control appear to allow?
- 2 Is the action discoverable without trial and error?
- 3 Does feedback confirm the action mode?

Hotkey labels do not look editable



OBSERVED ISSUE

I noticed that the shortcut combinations appear as static key labels. No Edit icon, instruction, or editable focus state is visible.

HCI IMPACT

Users may not discover that selecting the combination starts shortcut capture, increasing trial-and-error.

PROPOSED SOLUTION

Add a pencil/Edit control. On selection, show “Press a new shortcut” with Save, Cancel, and conflict feedback.

02

FINDING RELEVANT CONTROLS

Visibility

Can users find the control when they need it?

What visibility means

DEFINITION

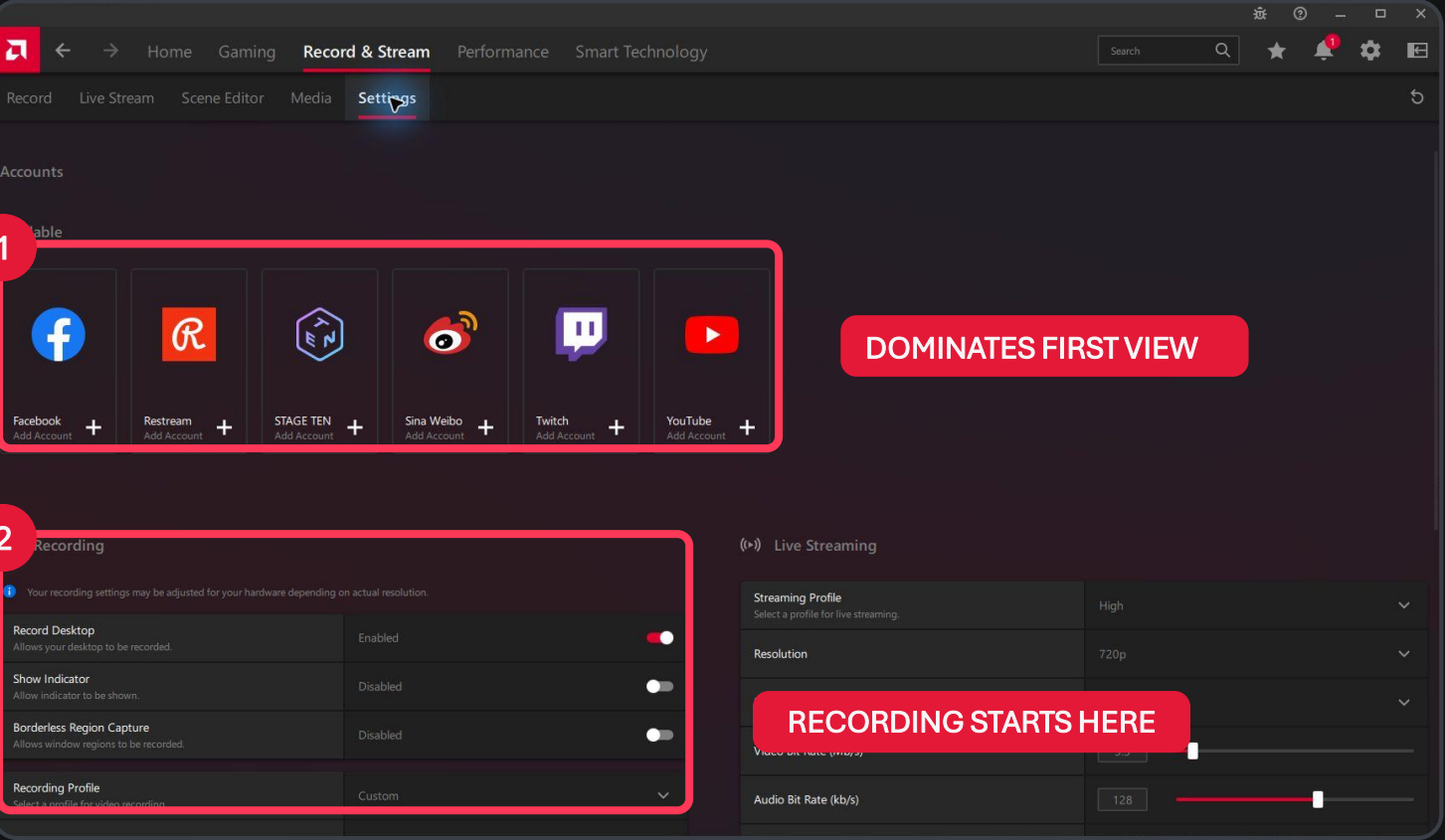
Visibility means that relevant controls, options and system states are apparent at the moment they are needed, without relying on memory or excessive searching.

High visibility reduces search cost and supports recognition over recall.

EVALUATION QUESTIONS

- 1 Is the primary task visible on arrival?
- 2 Do unrelated controls compete for attention?
- 3 Does the layout reveal what is below the fold?

Recording controls begin below streaming accounts



OBSERVED ISSUE

Streaming-account cards occupy most of the first view. Recording controls begin lower on the same long settings page. But there are multiple settings icons so it was difficult to find recording settings.

HCI IMPACT

A common capture task requires extra scanning and scrolling while unrelated streaming setup competes for attention.

PROPOSED SOLUTION

Use task tabs: Recording, Audio, Replay, Streaming. Open on the last-used section and place recording controls first.

03

PREDICTING CONSEQUENCES

Mapping

Does each control clearly predict its effect?

What mapping means

DEFINITION

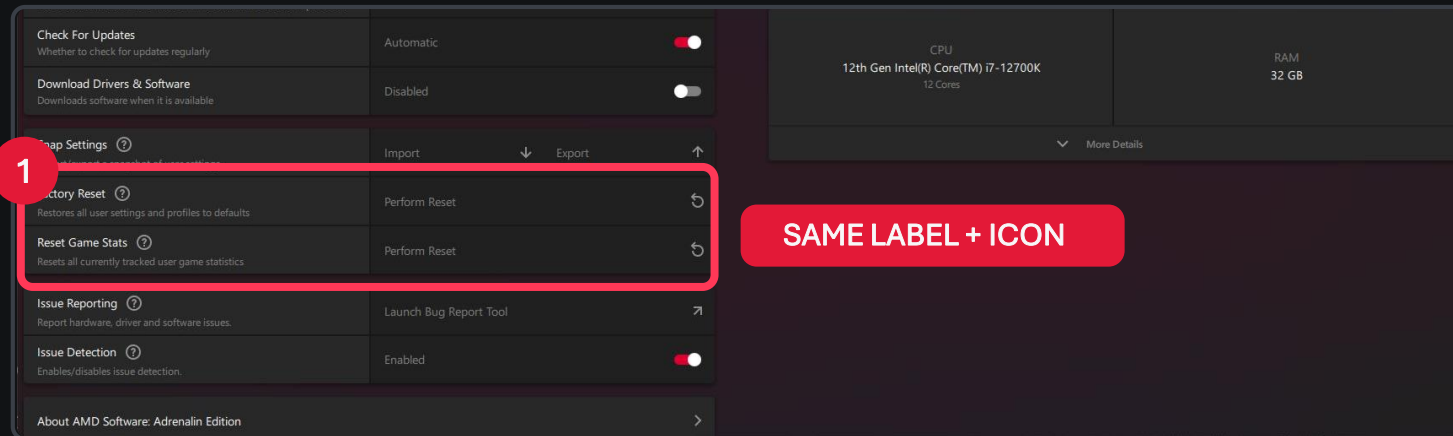
Mapping is the relationship between a control or action and its resulting effect. Good mapping lets users predict what will happen before they commit.

A control should communicate both the action and its scope.

EVALUATION QUESTIONS

- 1 Does the label match the outcome?
- 2 Are different consequences represented differently?
- 3 Is risk reflected in placement and styling?

Two identical reset controls imply similar consequences



OBSERVED ISSUE

Factory Reset and Reset Game Stats both use “Perform Reset” and both do the same things but one does extra stuffs. This can be confusing as they use same circular-arrow icon despite different scope.

HCI IMPACT

The mapping understates destructive consequence, creates hesitation, and raises the risk of choosing the wrong reset.

PROPOSED SOLUTION

Rename actions “Reset all AMD settings” and “Clear game statistics.” Isolate factory reset and confirm its scope.

04

BUILDING THE RIGHT MENTAL MODEL

System Image

Does the interface explain how the system works?

What system image means

DEFINITION

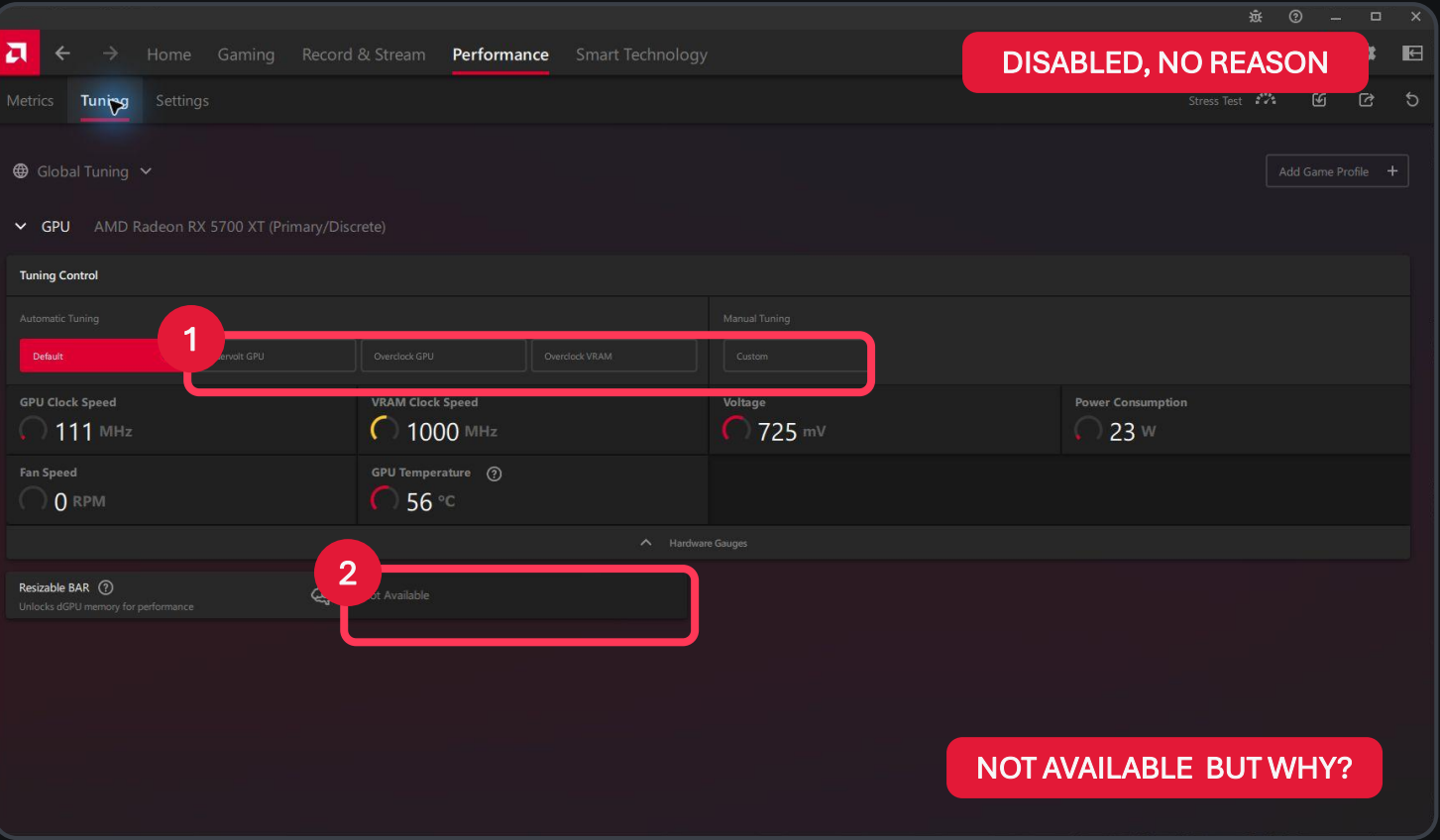
The system image is everything the user can observe like interface structure, labels, feedback, instructions and behavior through which the designer communicates a conceptual model.

When the system image is incomplete, users invent explanations.

EVALUATION QUESTIONS

- 1 Does the interface explain dependencies?
- 2 Can users infer why a feature is unavailable?
- 3 Does feedback support an accurate mental model?

Unavailable tuning options do not explain why



OBSERVED ISSUE

Tuning modes are disabled and Resizable BAR is “Not Available,” but the screen gives no visible reason or requirement.

HCI IMPACT

Users cannot form a correct model of whether hardware, firmware, driver, or configuration is responsible.

PROPOSED SOLUTION

Show reason text and a requirements checklist beside each disabled option, with a safe help or setup action.

05

BRIDGING ACTION AND FEEDBACK

The Gulfs

Can users act and interpret what happened?

A single screen is evaluated against this lens.

Two gaps can break one interaction

01 · BEFORE ACTION

Gulf of Execution

The gap between a user's goal and the actions they can identify and perform in the interface.

Question: What can I do next?



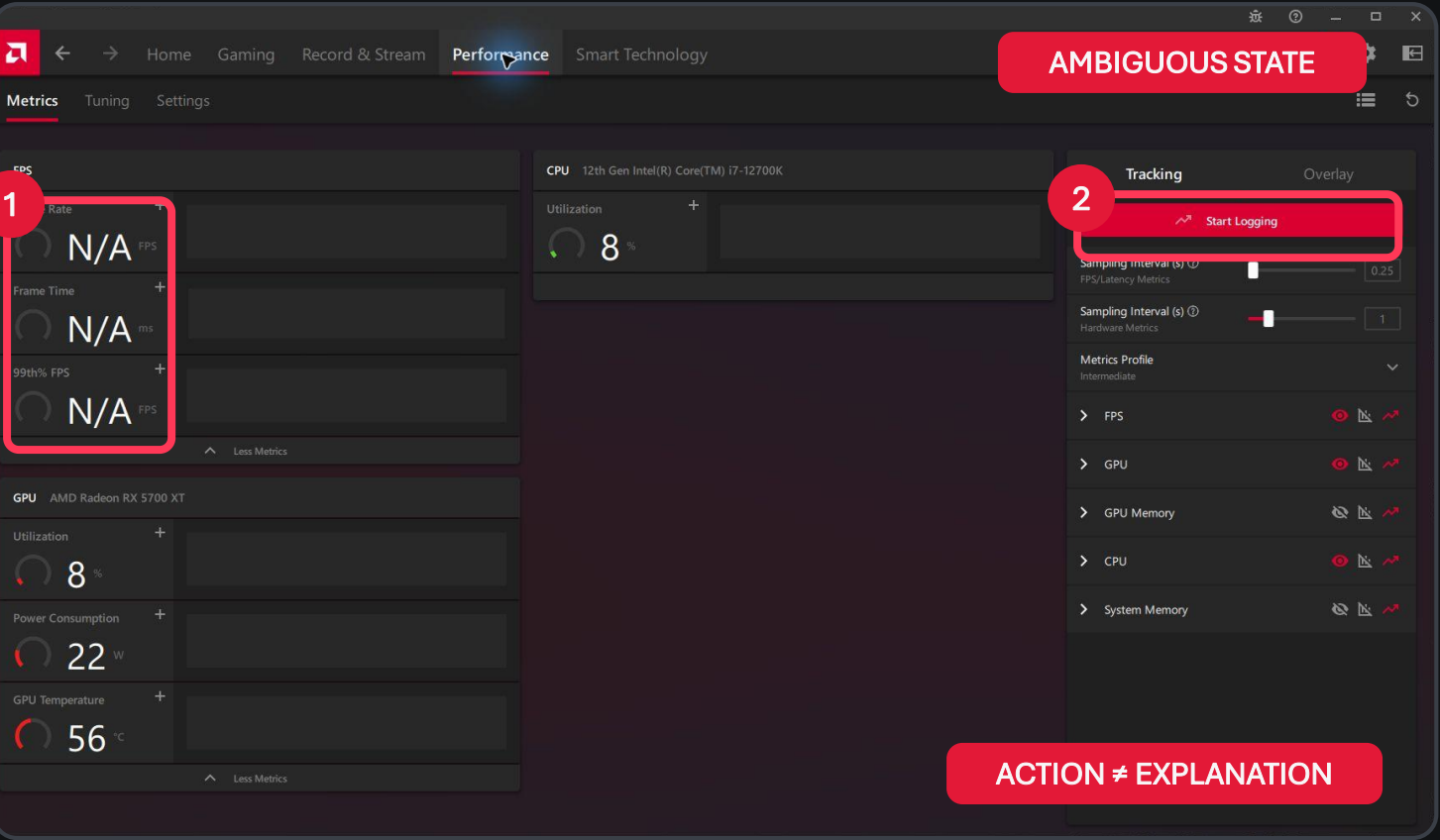
02 · AFTER ACTION

Gulf of Evaluation

The gap between the system's state or result and the user's ability to perceive and interpret it.

Question: What does this state mean?

FPS “N/A” gives no next step



OBSERVED ISSUE

Frame Rate, Frame Time and 99th% FPS show “N/A” while hardware metrics are live. No cause or enabling step is shown.

HCI IMPACT

Execution gulf: users cannot identify the next action. Evaluation gulf: they cannot tell unavailable from broken.

PROPOSED SOLUTION

Use state-specific guidance: “Launch a supported game to measure FPS.” Explain logging state, output, and errors.

Five changes would make AMD Software easier to control

01 Make edit states explicit

Add visible Edit cues, focus states, and shortcut conflict feedback.

02 Organize settings by task

Prioritize recording controls and separate recording from streaming setup.

03 Differentiate risky actions

Use action-specific reset labels and isolate destructive reset controls.

04 Explain unavailable features

Show why a control is disabled and what requirement is missing.

05 Replace N/A with guidance

Describe the current state and the exact next step to obtain a value.

The strongest redesigns make consequences, availability, and system state understandable.